

AMRITA VISHWA VIDYAPEETHAM
AMRITA SCHOOL OF COMPUTING, COIMBATORE
TIFAC-CORE IN CYBERSECURITY

20CYS384 Advanced Protocol Engineering Lab (L-T-P-C: 0-0-3-1)

Lab Evaluation 2 – IP Spoofing and DDoS Reflected with DNS, NTP and SSDP

Date: 06.03.2026

Program: B Tech CYS

Semester: 06

Name:

Duration: NA

Roll Number:

Maximum: 60 marks

To implement attacks using raw sockets.

Theory

- Raw Sockets: Bypass transport layer to craft custom packets.
- DDoS: Flood victim with reflected, amplified traffic from reputed server sources.

Scenario:

Write a Python script to spoof reflected flooding responses to a victim.

Procedure:

1. Use *scapy* to craft packets with specific details as required
2. Get amplified flooding packets sent to a target/victim from a trusted server and observe responses

Results Analysis:

Learn how to analyze program output using Wireshark or libpcap.

Expected Submission Deta

Zip/compressed file containing

1. **Python source code file** with comment and mnemonic variables
2. **requirements.txt** mentioning OS/environment, Python version and the version of Python modules you have used

3. **commands.txt** mentioning the python commands with command line argument and ip addresses for each attack
4. **3 separate pcap files named with the protocol** containing DNS, NTP and SSDP request packets (and response if you are able to manage it)
5. Word /PDF document – 3 sections for each protocol each containing
 - Format/structure of each protocol with the protocol data unit containing payload data you are creating at each layer using scapy and raw socket
 - Wireshark filter used to visualize the protocol request/response
 - Screenshot of the wireshark interface highlighting the data
 - Description explaining your output

Reference(s):

1. Tools:
 - a. Python's scapy
<https://scapy.readthedocs.io/en/latest/introduction.html>
<https://scapy.net/>
<https://github.com/secdev/scapy>
 - b. C's libpcap
2. Books:
 - a. Violent Python by TJ O'Connor

Evaluation Criteria:

The lab exercise will be assessed based on the following criteria:

- Completeness of the report. (20 marks)
- Originality and individual effort in completing the experiment. *If required, viva can be conducted to verify originality.* (10 marks)
- Accuracy of descriptions and results. (20 marks)
- Clarity of explanations and discussions. (10 marks)

Submission:

1. Document the procedure
2. Use any snipping tool to capture the output(s). Paste it in a word file (confirmation that you have individually tried). Describe what you observe and what you have understood in less than 10 sentences (roughly 200 words).
3. Save/Export/Convert the word document to PDF file format using following filename pattern:
`<roll_no>-lab<no>.zip`
E. g.
`cys23001-le2.zip`
4. Submit the documentation *on or before 06.03.2026: 12:00 MIDNIGHT IST.*
5. Mode of submission will be informed later via official WhatsApp group.